

Generation Interconnection System Impact Study Report

For Queue Project Q18-0935

Locust Ridge li Llc 138 kV

April 25, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-0935. The Interconnection Customer proposes a 177.3 MW battery energy storage facility interconnecting at the Locust Ridge li Llc 138 kV substation in the Pennsylvania area. The request was received on January 31, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 370 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$29,394,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ST NICHOLAS COGEN PROJECT-DREWS 138.0 kV	BOWEN STREET-RAWHIDE 138.0 kV L	124.8	157	20.3
TREGO-BEAVER 500.0 kV	ADAMS-ONANCOCK DIESEL GENERATION FACILITY 230.0 kV	115.6	4,242	9.5
WHEELABRATOR FRCKVILLE ENERGY 345/1 kV Auto	TALENERGY MARTINS CRK LLC HARWOOD SUBSTATION-KLINE TOWNSHIP COGEN FACILITY 345.0 kV	106.9	992	18.5
BOWEN STREET-WESTWOOD GENERATION LLC 138.0 kV	MORGANVILLE-WESTWOOD GENERATION LLC 345.0 kV	104.3	572	12.0
24TH ST-GRAY 138.0 kV	UPSCO-WIGWAM 345.0 kV	103.5	192	6.7
CARLTON HILLS TAP-SAND ROAD 138.0 kV	UPSCO-WIGWAM 345.0 kV	103.0	251	6.7
GONZALES-CANYON 138.0 kV	WIGWAM-UPSCO 345.0 kV	102.4	221	6.7
GONZALES SUBSTATION-24TH ST 138.0 kV	UPSCO-WIGWAM 345.0 kV L	101.8	204	6.7
KLINE TOWNSHIP COGEN FACILITY SUBSTATION 345-138 kV XFMR	UPSCO-WIGWAM 345.0 kV	101.8	512	5.1
SAND RD-CANYON 138.0 kV	WIGWAM-UPSCO 345.0 kV	101.7	228	6.7
LITTLE CHTE-FT HUMBUG 345.0 kV L	CORONET-FORT HUBUG 500.0 kV	101.5	1,720	5.4
WESTWOOD GENERATION LLC 345-138 kV Transformer	WESTWOOD GENERATION LLC-MORGANVILLE 345.0 kV line	101.5	486	10.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ST NICHOLAS COGEN PROJECT-DREWS 138.0 kV	Rebuild line with higher-capacity conductor	\$2,589,000	62% (\$1,594,000)
TREGO-BEAVER 500.0 kV	Rebuild line with higher-capacity conductor	\$262,513,000	2% (\$3,810,000)
WHEELABRTOR FRACKVILLE ENERGY 345/1 kV Autotransformer	Replace transformer with higher-rated unit	\$19,149,000	18% (\$3,380,000)
WESTWOOD GENERATIN LLC-BOWEN ST 138.0 kV	Reconductor line	\$4,940,000	67% (\$3,295,000)
24TH ST-GRAY 138.0 kV	Reconductor line	\$889,000	27% (\$238,000)
CARLTON HILLS TAP-SAND ROAD 138.0 kV	Reconductor line	\$4,286,000	27% (\$1,149,000)
GONZALES-CANYON 138.0 kV	Reconductor line	\$5,977,000	27% (\$1,602,000)
GONZALES SUBSTATION-24TH ST 138.0 kV	Reconductor line	\$2,057,000	27% (\$551,000)
KLINE TOWNSHIP COGEN FACILITY 345-138 kV Transformer	Replace transformer with higher-rated unit	\$11,393,000	22% (\$2,567,000)
SAND RD-CANYON 138.0 kV	Reconductor line	\$845,000	27% (\$226,000)
LITTLE CHTE-FT HUMBUG 345.0 kV L	Reconductor line	\$49,961,000	2% (\$827,000)
WESTWOOD GENERATION LLC 345-138 kV Auto	Replace transformer with higher-rated unit	\$14,969,000	68% (\$10,155,000)
	Total allocated to project		\$29,394,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.